

AI Engineer Roadmap

DSA runs parallel throughout | All other tracks are sequential

ALL SKILLS AT A GLANCE

Track A — DSA (Parallel, Throughout)

Arrays, Strings, LinkedLists, Recursion, Trees, Graphs, DP, Greedy, Tries, Segment Trees, Complexity Analysis, Competitive Programming

2. Deep Learning & Computer Vision

PyTorch, Autograd, CNNs, Transfer Learning (ResNet, EfficientNet), YOLO, MiDaS, OpenCV

4. Development Skills

FastAPI / Node.js+Express, REST APIs, React/Vite, SQL, MongoDB, Vector DBs, Git/GitHub

6. CS Fundamentals

OS (Scheduling, Threads, Deadlocks, Paging), DBMS (Normalization, ACID, Indexing, SQL), CN (OSI/TCP-IP, HTTP, TLS)

8. High-Level System Design (HLD)

Load Balancing, Caching (Redis), Message Queues (Kafka), Sharding, CAP Theorem, Rate Limiting

10. Placement Season Prep

CS Fundamentals Revision, Mock Interviews, Company-Specific DSA, Resume & Portfolio Polish

1. Classical Machine Learning

Scikit-Learn, Linear/Logistic Regression, Decision Trees, Random Forests, XGBoost, K-Means, PCA

3. Generative AI & NLP

Tokenization, Word2Vec, Transformers, LoRA/QLoRA, Hugging Face PEFT, LangChain, RAG, ChromaDB/Pinecone

5. Docker, Cloud & MLOps

Dockerfiles, AWS EC2, Model Serving, Monitoring, Retraining Pipelines

7. Low-Level Design (LLD)

SOLID Principles, Design Patterns (Factory, Singleton, Observer, Strategy), UML, Class Diagrams

9. Internships & Projects

GitHub Presence, 1 CV Project, 1 GenAI/RAG Project, Deployment, Internship, Open Source (optional)

How to read this roadmap: Track A (DSA) runs continuously in parallel with everything else, from Month 1 to placement season, it never pauses. Phases 1-10 are done **one at a time, in sequence**, finish one phase fully before starting the next.

Track A: DSA — Runs Parallel, Throughout

This track never stops. It runs alongside every phase below, from Month 1 until placement season ends.

Micro-Skills: Arrays, Strings, Linked Lists, Recursion, Trees, Graphs (BFS/DFS, Dijkstra, Kruskal, Topological Sort), Dynamic Programming (1D, 2D, Bitmasking), Greedy, Tries, Segment Trees, Time/Space Complexity analysis.

Free: Striver's A2Z DSA Course / takeUforward (YouTube & Website) • Codeforces (competitive practice)

Paid: LeetCode Premium (buy ~3 months before placements, for company-tagged questions)

Phase	Cadence
Months 1-6 (Basic + Advanced sheet)	Daily — 2 hrs weekday mornings, deeper sessions on free days
Months 7 onward (Maintenance)	2-3 problems/day + weekly spaced revisits of Pattern Bank
Last 3 months before placements	Shift to company-tagged questions via LeetCode Premium

Track B: Sequential Roadmap Overview

Everything below is done **one phase at a time**, in order. DSA (Track A) keeps running underneath all of these. Python and Math foundations (NumPy, Pandas, Linear Algebra, Calculus) are already completed and not repeated here.

#	Phase	Duration
1	Classical Machine Learning	2.5 months
2	Deep Learning & Computer Vision	2.5 months
3	Generative AI & NLP	2.5 months
4	Development Skills (Backend + Frontend + DB)	1.5 months
5	Docker, Cloud & MLOps	1.5 months
6	CS Fundamentals (OS, DBMS, CN)	3 months
7	Low-Level Design (LLD)	1.5 months
8	High-Level System Design (HLD)	2 months
9	Internships & Projects	Ongoing checkpoints
10	Placement Season Prep	1.5-2 months

Sequential Phases 1-10

Phase 1: Classical Machine Learning

Micro-Skills: Scikit-Learn API. Linear/Logistic Regression math. Decision Trees (Entropy/Gini impurity). Ensemble methods (Random Forests, XGBoost). Unsupervised learning (K-Means, PCA).

Free: StatQuest with Josh Starmer (YouTube) • Official scikit-learn documentation tutorials

Paid: ML Specialization by Andrew Ng / DeepLearning.AI (Coursera) • Python for Data Science & ML Bootcamp by Jose Portilla (Udemy, ~\$450 during sales)

Phase 2: Deep Learning & Computer Vision

Micro-Skills: PyTorch Tensors and Autograd. Custom nn.Module classes. CNNs (kernels, padding, strides). Transfer learning (ResNet, EfficientNet). Vision: YOLO prediction heads, MiDaS depth tensors, live webcam feeds with OpenCV.

Free: PyTorch for Deep Learning: Zero to Mastery by Daniel Bourke (YouTube) • Fast.ai — Practical Deep Learning for Coders

Paid: Deep Learning Specialization by Andrew Ng (Coursera)

Milestone: build 1 Computer Vision project by the end of this phase.

Phase 3: Generative AI & NLP

Micro-Skills: Tokenization, Word2Vec. Transformer Architecture (Query, Key, Value vectors). Fine-tuning (LoRA, QLoRA using Hugging Face PEFT). RAG pipelines (LangChain, chunking, ChromaDB/Pinecone).

Free: Hugging Face NLP Course (official, free) • Neural Networks: Zero to Hero by Andrej Karpathy (YouTube)

Paid: Generative AI with LLMs by DeepLearning.AI (Coursera)

Milestone: build 1 GenAI/RAG project by the end of this phase.

Phase 4: Development Skills

Micro-Skills: REST APIs with FastAPI (Python) or Node.js/Express. React/Vite basics for simple UIs. SQL (Postgres/MySQL) + MongoDB aggregations. One vector database (ChromaDB or Pinecone). Git/GitHub — branching, PRs.

Free: freeCodeCamp (YouTube: Node.js and React full courses) • FastAPI Official Documentation

Paid: — well covered by free resources

Phase 5: Docker, Cloud & MLOps

Micro-Skills: Writing Dockerfiles. Deploying containers to AWS EC2. Model serving, monitoring, and basic retraining pipelines.

Free: FastAPI Official Documentation • AWS Free Tier docs + hands-on EC2 practice

Paid: Machine Learning Engineering for Production (MLOps) Specialization (Coursera)

Milestone: deploy both projects (CV + GenAI) using Docker + EC2 by the end of this phase.

Phase 6: CS Fundamentals

6.1 OS: CPU Scheduling (RR, SJF), Multithreading, Mutexes, Semaphores, Deadlock handling (Banker's Algorithm), Paging, segmentation, thrashing.

Free: Neso Academy / Gate Smashers • Paid: Grokking the OS Interview (Educative.io)

6.2 DBMS: Relational algebra, Normalization (1NF-BCNF), ACID Properties, B-Trees/B+ Trees, Advanced SQL (window functions, joins).

Free: CMU DB Group — Andy Pavlo (YouTube) • Paid: Complete SQL Bootcamp by Jose Portilla (Udemy)

6.3 CN: OSI vs TCP/IP, TCP vs UDP, Subnetting, HTTP/1.1 vs HTTP/2 vs WebSockets, SSL/TLS.

Free: Neso Academy — CN Playlist • Paid: Kurose & Ross — Computer Networking (book)

Phase 7: Low-Level Design (LLD)

Micro-Skills: SOLID Principles. Design Patterns (Factory, Singleton, Observer, Strategy). Class diagrams and UML. Writing maintainable, extensible code.

Free: Refactoring.guru (Design Patterns) • Christopher Okhravi — Design Patterns playlist (YouTube)

Paid: Grokking the Object-Oriented Design Interview (Educative.io)

Phase 8: High-Level System Design (HLD)

Micro-Skills: Load Balancing (Nginx, HAProxy). Caching (Redis, Memcached). Message Queues & Event Streaming (Kafka, RabbitMQ). Database Sharding and CAP Theorem. Rate limiting.

Free: Gaurav Sen — System Design Playlist (YouTube) • ByteByteGo (YouTube, Alex Xu)

Paid: System Design Interview Vol 1 & 2 by Alex Xu (books — top priority buy) • ByteByteGo Premium Course

Phase 9: Internships & Projects — Checkpoints

Not a separate time phase — these are checkpoints to hit alongside the phases above.

Checkpoint	Target
GitHub presence started	Early — during Phase 1
1st project (Computer Vision) built	End of Phase 2
Internship applications begin	Start of Phase 3
2nd project (GenAI/RAG) built	End of Phase 3
Both projects deployed (Docker + AWS)	End of Phase 5
Internship completed (target)	Before Placement Season
Resume + portfolio finalized	Start of Phase 10

Optional differentiators: Competitive Programming rating on Codeforces, Open Source contributions once comfortable with a framework (e.g. LangChain, Hugging Face).

Phase 10: Placement Season Prep

Micro-Skills: Full CS Fundamentals revision (crash mode). Mock interviews — DSA, ML/system design, behavioral. Company-specific DSA practice using LeetCode Premium. Final resume and portfolio polish.

Free: Mock interview practice with peers/seniors • Company-specific question banks (LeetCode Premium)

Paid: Interview prep courses/platforms as needed closer to the date